

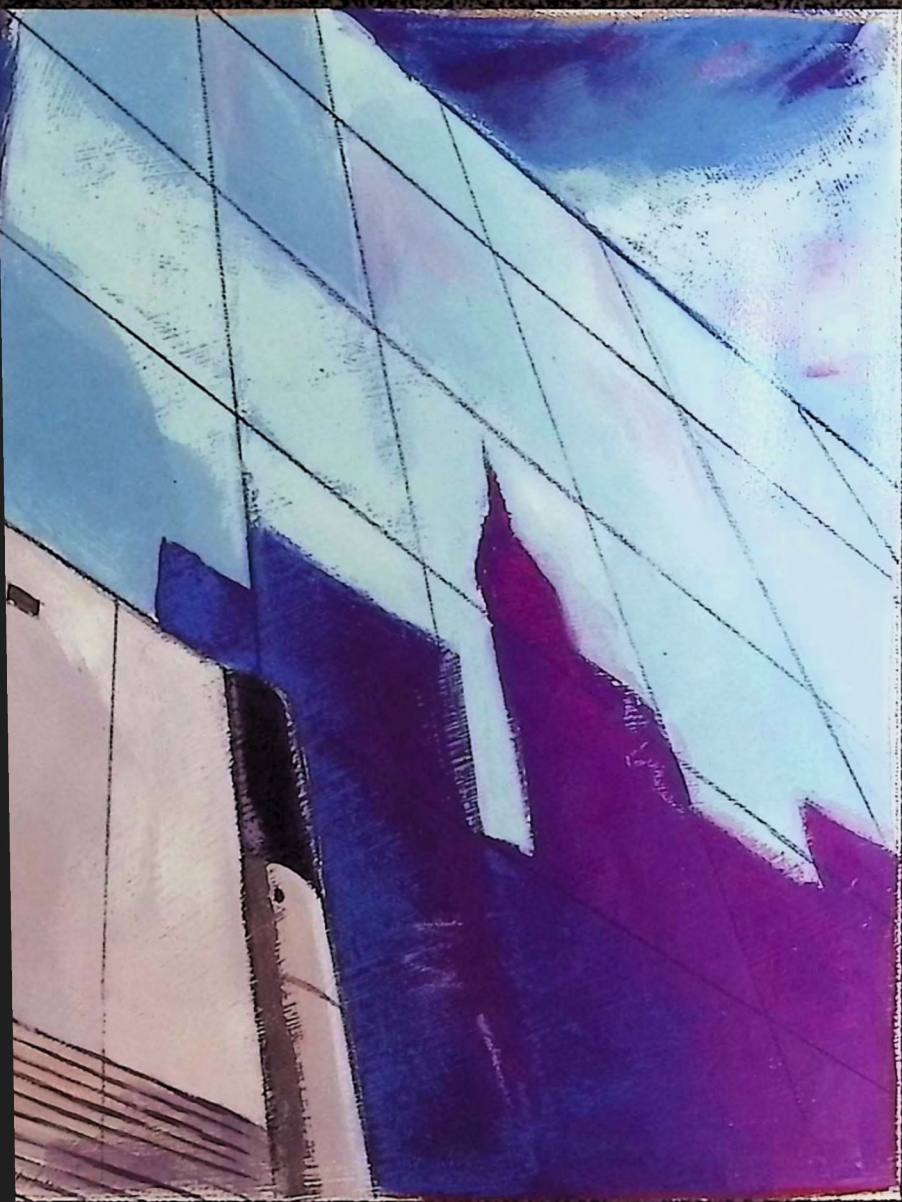


Model 3550

NCR
System
3000



Open,
Cooperative
Computing.
The Strategy
for Managing
Change.



Today, open information processing systems provide unique opportunities for your business to increase its competitive edge.

Through transparent communication links between people and information, your business can provide efficient service while at the same time cutting costs and facilitating better day-to-day decisions.

NCR System 3000 is designed to offer the broadest range of open, scalable systems in the computer industry today. From
NCR System 3000 can help you
manage changes in technology
uniprocessor to multiprocessor systems and from tightly-coupled to
as well as end-user and
loosely-coupled architectures, the NCR System 3000 product
organizational needs. A line of
strategy helps you economically match system performance to
scalable systems lets you add
current business needs. What's more, with any system you choose,
processing power in increments,
you have the flexibility to adapt and expand it as your needs
with minimal cost and disruption.
evolve.

The NCR 3550 is the most powerful tightly-coupled system in the System 3000 family. Designed to deliver several times the performance traditionally found in mainframe computers, for less cost, the NCR 3550 lets you incorporate an open system as the cornerstone of your enterprise information source.

Combining industry standard I/O components with NCR's dual system bus architecture, the NCR 3550 draws upon the research and development investment of the entire computer industry.

NCR's goal? To deliver an affordable, high performance, reliable, scalable system.



Performance

Fast information access is critical to meeting your enterprise application needs. To meet this demand, the NCR 3550 combines several key features to create a system capable of processing over 250 transactions per second in Relational Database Management System environments. These features merge to form a system capable of supporting your information processing strategy with advanced technology and power.

Tightly-Coupled Multiprocessing

The NCR 3550 is a tightly-coupled, fully symmetric multiprocessing system linking two to eight 50MHz Intel486™ processors, each operating at 40 MIPs for a total of 320 MIPs. Each processor in the symmetrical architecture is supported by 256K of cache, can perform any function, and can take full advantage of multiprocessing capabilities.

The tightly-coupled, symmetric architecture of the NCR 3550 provides compatibility with System 3000 uniprocessing applications; there is generally no need to modify applications to run on this multiprocessing system.

Dual Bus Architecture

The key to this tightly-coupled system is NCR's highly advanced, dual processor-to-memory bus design. With two 64-bit buses operating at 25MHz, this system provides peak throughput of 200 Megabytes (MB) per second per bus, for a total of 400MB per second. Sustained throughput is 113MB per second per bus, with a total sustained throughput of 226MB per second. The parallel paths between the processors and the memory help decrease bus contention because the processors are divided across the two buses.

Other designs using more processors may exceed the capacity of the bus, leading to increased bus contention and degraded performance. The NCR 3550 is designed to give you full processing power by minimizing bus contention.

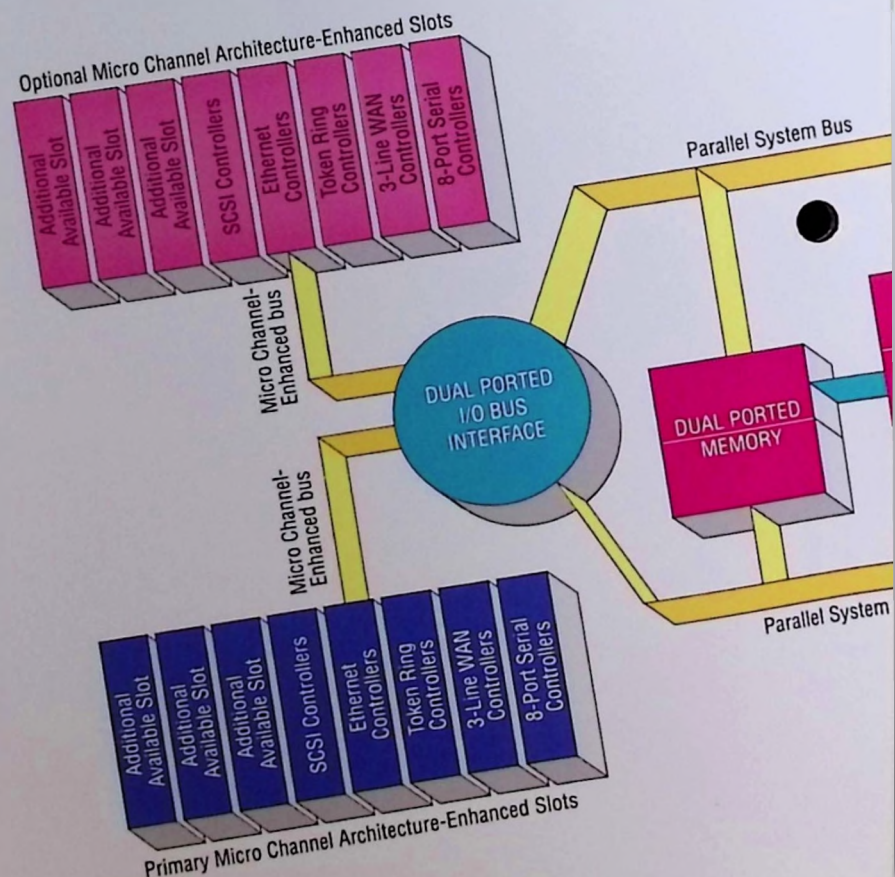
This balance of bandwidth and power gives you a system designed to provide optimal performance.

Dual-Ported, Four-Way Interleaved Memory

The system's memory design distributes all memory addresses equally across as many as four memory interleaves. With dual-ported, four-way interleaved memory, the NCR 3550 can have two processors, each accessing up to four different memory locations simultaneously.

Directory-Based Cache Coherency

Maintaining cache coherency between multiple processors always requires some overhead. This overhead is incurred as processors check the bus for memory location addresses before reconciling the cache and memory with new data. With a "bus snooping" method of insuring coherency, the processors check the bus periodically, whether or not another processor needs the memory location.



The NCR 3550, however, uses a directory-based cache coherency method designed to minimize this overhead. This scheme uses a directory in memory to mark the status of a memory line. When a processor writes to cache, it also changes the status of the memory location. When another processor needs the same memory location, it checks the status of memory. If it's not current, the logic in memory accesses cache and retrieves the new data to ensure its coherency. As a result, overhead is only incurred when one processor needs the same memory address written by another processor.

I/O Subsystem

While superior processor performance is critical for a high performance system, the I/O subsystem must be equally tuned. NCR selected the industry standard

Micro Channel Architecture[®]-Enhanced bus as the architecture for the multiprocessing members of the System 3000 family. This bus is designed to offer the highest performance of any commercially accepted open system I/O bus. In addition, the I/O bus supports data integrity and auto-configuration, requiring no strapping options.

With an I/O rated bandwidth of 80 MB/second, the Micro Channel Architecture-Enhanced bus provides support for all I/O devices, including SCSI, LANs, WANs, and terminals. The NCR 3550 can be configured to have two Micro Channel Architecture-Enhanced buses. Each Micro Channel Architecture-Enhanced I/O bus has access to both system buses and, like the memory, is dual ported, thus providing equal access to all CPUs. The NCR 3550 offers up to 16 Micro Channel Architecture-Enhanced slots, giving room to grow and add functionality as needed.

The NCR 3550 design also includes up to eight Dual Channel SCSI Controllers providing two SCSI channels each, for a total of 16 SCSI channels for the system. Supporting SCSI II protocols, each channel can connect up to seven storage devices.

Storage Devices

Because the NCR 3550 has such a powerful, versatile I/O subsystem, the system is designed to offer all of the advantages that come with:

- Multiple individual drives
- High-speed, 5400-RPM hard disk devices
- A distributed file system.

Drives

The NCR 3550 supports up to twenty eight 5.25" full-height fixed drives and four full-height removable media devices. Alternately, each bay may support two 3.5" devices with removable bays supporting dual half-height 5.25" drives.

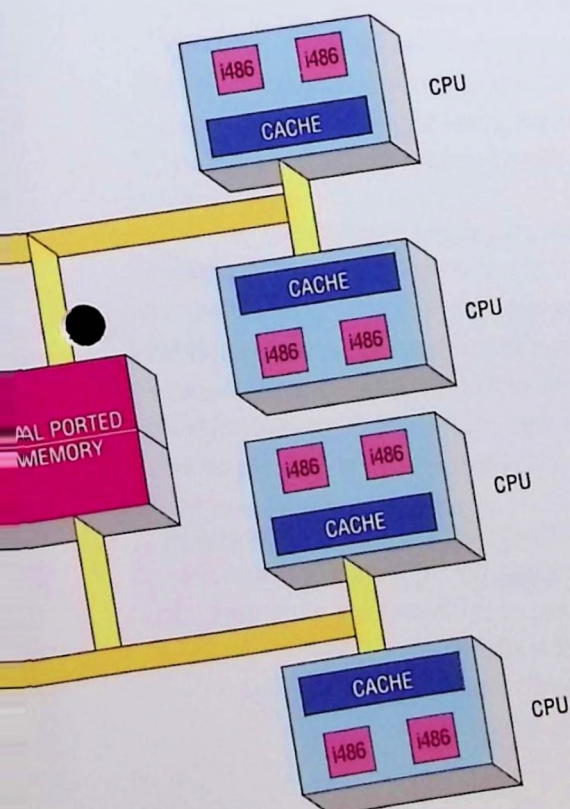
The NCR 3550 allows you to distribute data over a number of disks. This capability is designed to decrease the effect of seek time and rotational latency, thus providing faster data access. This is achieved by simultaneously accessing multiple pieces of data from the same file system.

High-Speed Devices

To maximize the performance of the large number of drives and channels available, the NCR 3550 supports 5400-RPM devices that help reduce disk rotational delay. As a result, the transfer rate increases because the disk moves faster under the heads.

Distributed File System

Balancing the I/O load ensures that you get the most efficient performance from your system devices. The NCR 3550's distributed file system works to ensure that all storage devices are used equally to maintain an efficient level of storage and retrieval.



NCR UNIX® System V Release 4 Operating System

The NCR UNIX System V Release 4 (SVR4) operating system is designed to offer a significant competitive edge. This operating system scales the entire System 3000 family taking full advantage of both uniprocessor and multiprocessor hardware designs. The NCR UNIX System V Release 4 operating system is a fully symmetrical implementation for multiprocessing systems such as the NCR 3550.

This allows any processor to execute any combination of kernel and user processes or I/O requests, virtually eliminating potential software bottlenecks. In addition, the kernel is multi-threaded, meaning that multiple parts of it can be executed in parallel.

Based on requirements defined by UNIX International, SVR4 combines the functionality of UNIX System V, BSD and XENIX® and complies with major standards, including POSIX, XPG3, FIPS 151-1, ANSI C, iABI and iBCS-2 to ensure interoperability. In addition, internal interfaces, such as device drivers and STREAMS modules, which support multiprocessing capabilities, as well as user-visible Application Programming Interfaces (APIs), are supported.

Another outstanding feature of the NCR operating system is asynchronous I/O. By permitting applications to initiate I/O requests and continue processing while the I/O request completes, this operating system allows applications to overlap I/O requests, helping to improve system throughput.



Availability

In addition to superior performance, the NCR 3550 is built to provide the system availability critical for enterprise applications and mainframe-class processing. Implementing high-availability features on top of a reliable hardware platform, NCR presents a system that is designed to meet the levels of availability required by enterprise applications, without redundant components which duplicate work. NCR calls this concept "fault resilient".

NCR has included in the NCR 3550 such features as error detection and correction, fault-tolerant features in critical areas and on-line service, further contributing to system availability.

To help ensure high reliability, NCR strives to use only the highest quality components, conducts extensive system design simulations and requires stringent system testing, including environmental conditions tolerance testing.

These characteristics are aimed at helping the NCR 3550 achieve a level of system availability and reliability that approaches that of fault-tolerant systems but at a much lower cost.

Error Detection and Correction

To detect, isolate and correct any system faults that may occur, the NCR 3550 incorporates a dedicated Diagnostic Processor, Built-in Self Test and Boundary Scan, and Error Check and Correct Logic on system memory. Contributing further to the ability to diagnose and correct system errors, the NCR 3550 includes parity on data and addresses on system, Micro Channel Architecture-Enhanced and SCSI buses.

Fault-Tolerant Features

Lost or corrupted data means lost time and money. To help protect against such losses, the NCR 3550 includes fault-tolerant features in areas critical to maintaining data integrity. Power Fail Recovery, Disk Mirroring, Redundant Arrays of Inexpensive Disks (RAIDs) and SCSI controller redundancy are designed to provide this protection.

Power Fail Recovery

The NCR 3550 power-backup system helps ensure data integrity in the event of intermittent or total power outages. This system protects against intermittent power losses by keeping the system running on battery power. In the event of a total power outage, the batteries will then retain the memory content intact until it can be transferred to disk. Once the power is restored, the system automatically loads the memory from the disk so you can continue processing.

Redundant Arrays of Inexpensive Disks

Disk mirroring requires that for each disk, there is a back-up disk. While this method offers a high level of data protection, RAID's are an alternate and generally a more cost effective method.

Disk arrays stripe the data files across multiple disk drives while generating and storing parity information on these files. This parity information typically requires only 25% additional storage space, as compared to the 100% redundancy required for mirroring. If one disk fails, the array controller can reconstruct the data by utilizing the parity information along with the data stored on the remaining drives in the array.

Disk Mirroring

Some users prefer the assurance of a one-to-one disk relationship for near 100% protection. For these users' critical data protection needs, disk mirroring may provide the answer. Each time there is a write to disk, a backup disk gets the same information written to it—the

two disks "mirror" each other. As a result, loss of a disk does not mean the loss of data.

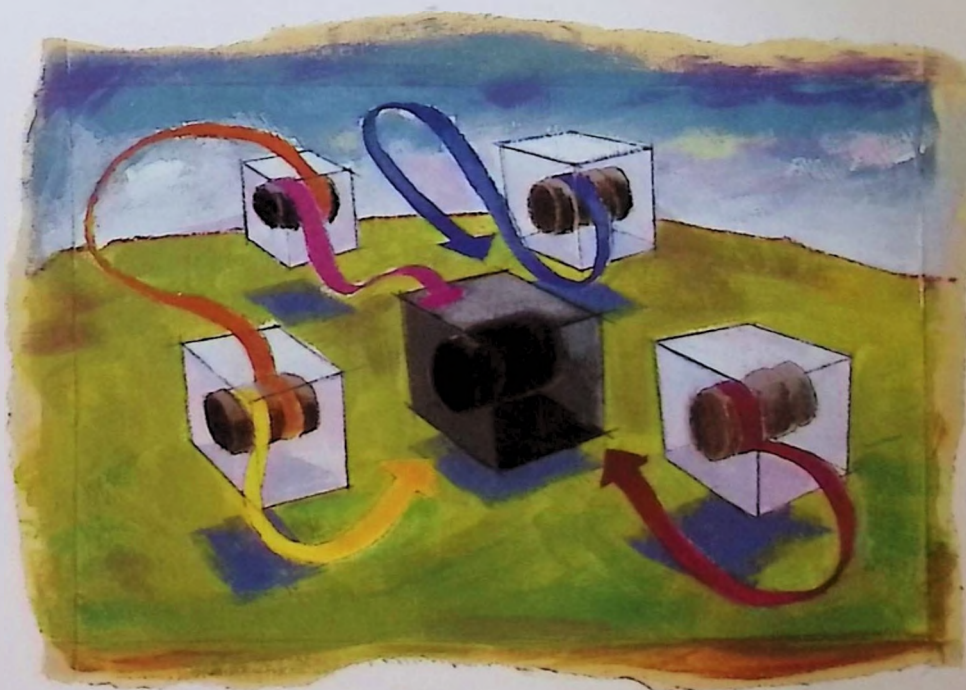
SCSI Controller Redundancy

To help protect the data flow from media devices to system memory, the NCR 3550 offers SCSI controller redundancy. Because of this available controller backup, if one controller fails, the system can continue meeting your business needs.

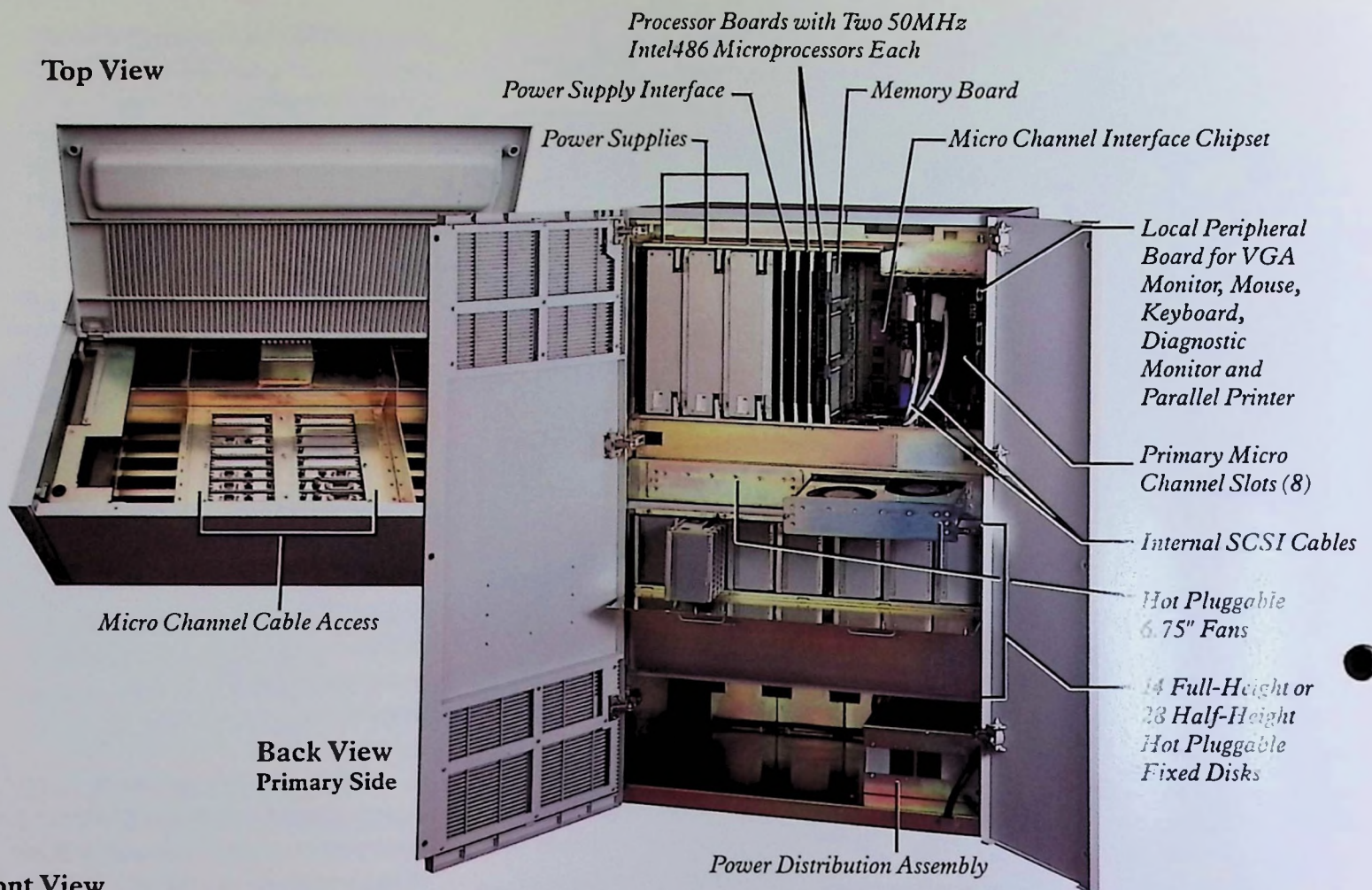
On-Line Service

Hot pluggable drives allow system maintenance while the system is on-line, helping to save you valuable processing time. You can stay operational while the disk is replaced and you can keep the system on-line to restore the data. In addition, the system console uses a series of menus to explain the repair procedure if you choose to perform your own disk service.

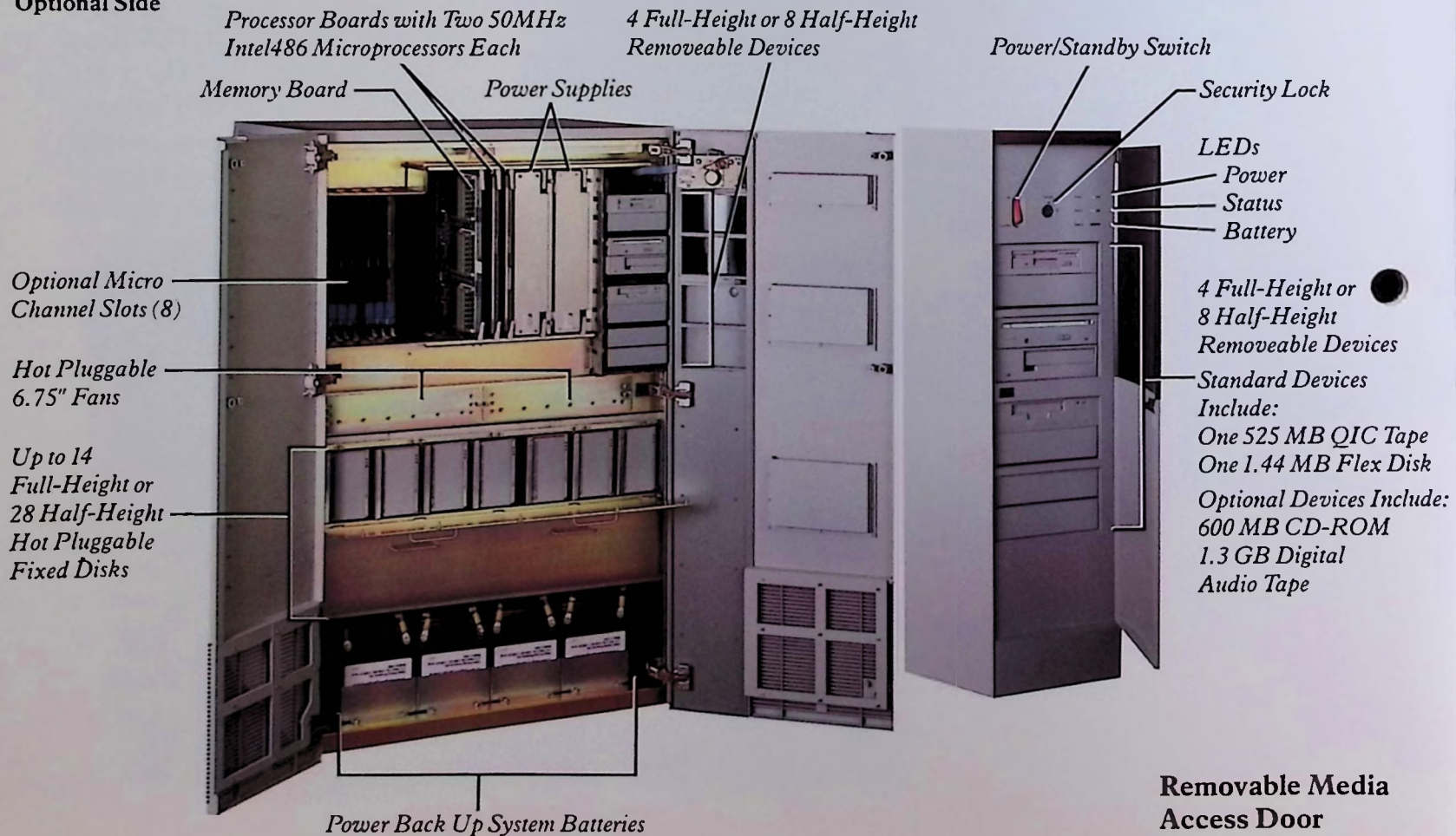
This same technology allows fans used to cool the system to be replaced while the system is operational.



Top View



Front View Optional Side



Scalability

Clearly, the NCR 3550 is designed to offer the superior performance and system availability needed in the enterprise environment.

However, these qualities must be accompanied by growth potential to meet the future requirements of a growing enterprise system. The NCR 3550 rises to this challenge as well.

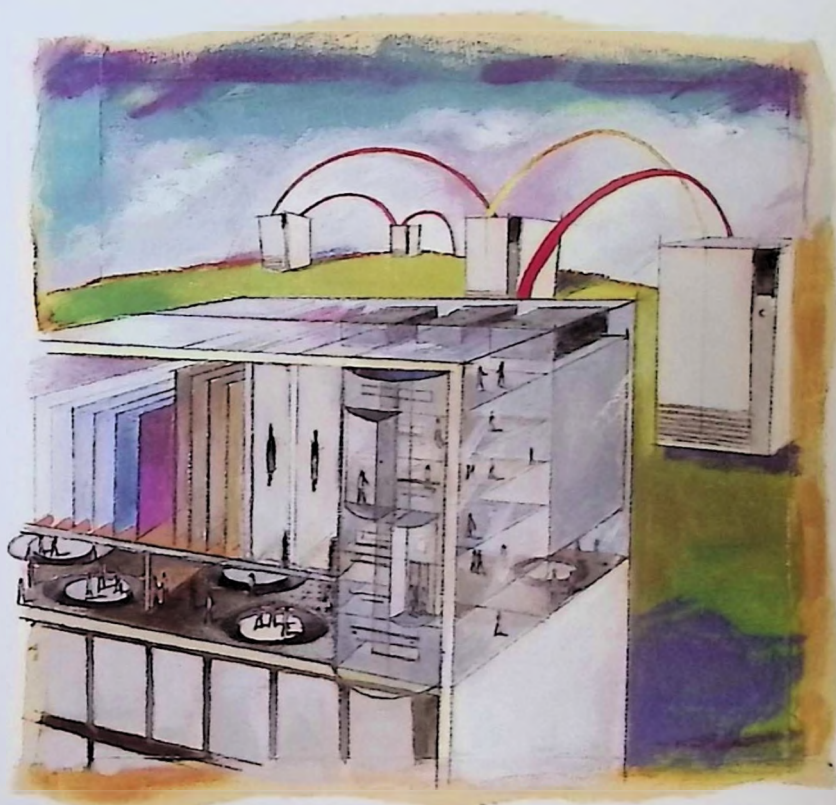
System growth potential stems from outstanding feature flexibility:

- 2 to 8 processors (80 to 320 MIPs)
- 32 to 512MB memory
- 8 or 16 Micro Channel Architecture-Enhanced I/O slots
- 340MB to 36GB internal disk storage
- Up to 100GB external storage capacity
- 2 to 16 SCSI channels
- 1 to 4 Ethernet LAN channels
- 1 to 4 Token Ring connections
- 1 to 4 V.35 WAN connections
- 1 to 8 RS-232 WAN connections

The NCR 3550 gives you a solid growth path. Depending on your needs, you can begin with a two-processor system and, as demand dictates, grow to an eight-processor system supporting up to 1024 users — all within a single cabinet.

In addition to providing you with outstanding processing power, speed and reliability, this system gives you the means to match memory, disk and other peripheral options to processor growth.

Supporting the strategic Network File System Operating Environment and LAN Manager for UNIX as offered throughout the System 3000 family, the NCR 3550 allows software to migrate and scale the family. Clearly, the NCR 3550 is designed to help you incrementally and economically expand your system to coincide with your business' progress.



Formula for Success

NCR System 3000 is built to offer the broadest range of enterprise systems in the industry. As part of the System 3000 family, the NCR 3550 is designed to provide superior reliability, high availability and system scalability to support your mission-critical enterprise applications. We think you'll agree the NCR 3550 meets today's information processing demands and offers a clear, cost-effective direction for growth tomorrow.



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